

Fournier's Gangrene: Contemporary Approach to Treatment and Outcomes

Learning Objective: At the conclusion of this continuing medical education activity, the participant will be able to identify risk factors associated with the development of Fournier's gangrene and recognize the approach to diagnosis, acute surgical management, and subsequent reconstruction.

This AUA Update aligns with the American Board of Urology Module on Core/General Urology. Additional information on this topic can be found in the AUA Core Curriculum sections on Reconstructive Urology and Urologic Infections.



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KEY WORDS: Fournier's gangrene, necrotizing soft-tissue infections of the genitalia, surgical debridement, genitourinary reconstruction, perineum

INTRODUCTION

Fournier's gangrene (FG) is a rare necrotizing fasciitis of the perineal, genital, and perianal regions. While it was first described nearly a hundred years prior by Baurienne,¹ it was named after a French venereologist, Jean-Alfred Fournier, after he presented a 5-case series titled "Gangrène foudroyante de la verge" in 1883.² In this series he described a rapidly progressive necrotizing fasciitis of the genitalia. All patients in this series suffered from what is now known as FG.²

Once thought to be idiopathic in origin, numerous risk factors have now been associated with the development of FG, including diabetes, chronic alcoholism, immunodeficiency, chronic steroid use, oncologic conditions, cytotoxic drugs, malnutrition, and low socioeconomic status (SES).^{3,4} And while FG has a male preponderance of 10:1, it can affect women and children.⁵

Treatment of FG entails rapid diagnosis, antibiotic therapy, and debridement. Once the patient is stabilized, reconstructive options to restore the remaining defects are then prioritized. It is estimated that up to 67% of patients will need some degree of reconstruction afterward.⁶

PATIENT DEMOGRAPHICS

FG predominantly affects elderly males with an incidence of 1.6 in 100,000 males in the United States, with peak incidence in those ages 50 to 79 years old.⁷ Low SES has been associated with an increased likelihood of an individual developing FG, thought to be due to the higher rates of diabetes and alcoholism in these populations, which are often risk factors.⁸ Akilov et al found that the majority of patients who developed FG were of low-income households, defined as an annual income of <\$16,400 for a family of 4.⁹ This suggests that the most vulnerable population affected by FG is those with poor access to adequate health care, a concerning reality given that delayed presentation and extended time to debridement has been associated with increased mortality rate.¹⁰

PATHOPHYSIOLOGY

Necrotizing soft-tissue infections of the genitalia trigger an inflammatory response resulting in cell death that further progresses to obliterative endarteritis with microthromboses along fascial planes.¹¹ **Often originating at the external genitalia or perineum, FG can progress along dartos fascia (involving the penis and scrotum), Colles' fascia (involving the perineum), and even Scarpa's fascia (involving the abdominal wall), which is contiguous with Colles' fascia.**¹² Resulting edema and compromised blood supply cause progressive exponential perifascial dissection with overlying skin and subcutaneous tissue necrosis.³ Fascial

necrosis can occur at an alarming rate of up to 2-3 cm per hour—further illustrating the need for rapid diagnosis and treatment.¹³ **Although in close proximity to the sites of necrosis, testicular involvement in FG is not often seen—credited to the separate blood supply of the testicles by the gonadal arteries.**¹³ In some cases, however, orchiectomy is necessary. Overall, if FG is allowed to progress, systemic involvement is imminent, including sepsis, multisystem organ failure, and death.

MICROBIOLOGY

Previously thought to be exclusively due to streptococcal infections, FG is now recognized to be a polymicrobial phenomenon.¹⁴ Furthermore, it is believed to be an infection that progresses synergistically. Alone, the inciting bacteria likely would not result in the fulminant progressive nature of FG. Most believe that the polymicrobial involvement is necessary to create the synergy of enzyme production that promotes rapid multiplication and spread of FG.⁵ For example, one microorganism might produce the enzymes necessary to cause coagulation of the nutrient vessels. Thrombosis of these nutrient vessels reduces local blood supply; thus, tissue oxygen tension falls. The resultant tissue hypoxia allows for the growth of facultative anaerobes and microaerophilic organisms. These latter microorganisms, in turn, may produce enzymes (eg, lecithinase, collagenase) which lead to digestion of fascial barriers, thus fueling the rapid extension of the infection.^{5,15}

A recent review identified a total of 1,227 patients with culture-proven polymicrobial or monomicrobial infectious etiologies of FG. Polymicrobial infections were identified in 58.4% of patients and monomicrobial in only 30.1% of cases. The most causative organisms were noted to be *Escherichia coli* (26.6%), *Staphylococcus* spp (13.8%), *Streptococcus* spp (11.3%), and *Pseudomonas* spp (8.6%). Sterile cultures are also possible and were found in 18.7% of cases.¹⁶

With the ever-growing concern for multidrug-resistant organisms (MDROs) resulting in genitourinary tract infections, there is concern that MDROs are an increasing cause of FG. Chia and Crum-Cianflone identified 12 patients in their cohort who had FG caused by MDROs, the majority of which were methicillin-resistant *Staphylococcus aureus*.¹⁷ Interestingly, MDROs became more prevalent as their study progressed, with 67% of FG cases attributable to MDROs in the final 3 years of their 10-year study. Immunosuppression (caused by diseases such as diabetes, cancer, chronic alcoholism, HIV/AIDS, renal failure, etc¹⁶) and chronic wounds were associated with MDROs and may indicate a patient population that should be targeted with aggressive empirical antibiotic therapy.¹⁷

RISK FACTORS

Although originally described by Fournier as an idiopathic infection, FG has an identifiable cause in up to 95% of cases.¹⁸ These necrotizing soft-tissue infections of the genitalia often

ABBREVIATIONS: Fournier's gangrene (FG), Fournier's Gangrene Severity Index (FGSI), hyperbaric oxygen therapy (HBOT), Laboratory Risk Indicator for Necrotizing Fasciitis (LRINEC), multidrug-resistant organism (MDRO), negative-pressure wound therapy (NPWT), socioeconomic status (SES), vacuum-assisted closure (VAC)

originate from local trauma to the genitourinary tract, or skin of the genitalia or anorectum, thus allowing entry of bacteria into the soft tissues. Examples of genitourinary causes of FG include prostatic abscess, urethral stricture, urethral stones, and renal abscesses.¹⁹ Anorectal causes of FG include colorectal injury or a complication of colorectal malignancy, inflammatory bowel disease, diverticulitis, and appendicitis.²⁰ In a recent practical review, skin pathology was responsible for FG in 24.3% of patients, urological causes were the source in 16.8%, gastrointestinal sources accounted for 11.9%, traumatic injury was the source in 5.1%, and 32.4% of patients had unclear etiology.¹⁶

Any comorbidity that results in immunocompromise may predispose a patient to the development of FG. Most common of these comorbidities is diabetes, which was found to be a predisposing factor in 32% of cases.¹⁶ Chalya et al found patients with diabetes to have higher mortality.²¹ Furthermore, patients with hemoglobin A1c levels greater than 7% were found to have an increase in length of hospital stay and mortality.²² Chronic renal failure has also been associated with higher mortality in patients with FG.²³

As previously stated, there may be an association of higher rates of FG in those with low SES.⁹ Certain barriers such as access to health care and transportation are likely barriers to patients most at risk for FG, which hinder their ability to present for medical care in a timely fashion.

CLINICAL PRESENTATION AND DIAGNOSIS

The diagnosis of FG is often a clinical one. Diagnosis can be difficult due to the vague and nonspecific symptoms that patients present with. **It is important to have FG as a differential diagnosis early on, as prompt diagnosis leads to earlier debridement, which has been found to have a mortality benefit.** Physical examination must include evaluation of the abdomen, perineum, genitalia, and rectum to determine the extent of involvement. Early findings may include edema, erythema, fluctuance, and purulent drainage. Crepitus is often thought to be pathognomonic for FG in the correct clinical setting, although this depends on the specific bacteria that is causing the underlying infection.

Although not always necessary, imaging can be an important diagnostic tool that can point a clinician to the diagnosis of FG. Ultrasound can be used to detect subcutaneous emphysema, underlying scrotal abscess, as well as help exclude other diagnoses such as epididymo-orchitis or testicular torsion.²⁴ CT is another useful modality that can quantify extent of disease progression and abdominal involvement. Perry et al found that CT scan aided in making a diagnosis by showing underlying gas, persistent abscess, and inflammation with fat stranding.²⁵ CT may also be useful in determining extent of disease, which, in turn, may guide surgical planning. Urgent diagnosis of FG is imperative, so if there is high clinical suspicion, imaging should not delay operative intervention.

The Laboratory Risk Indicator for Necrotizing Fasciitis (LRINEC) has become a well-established tool for aiding in the diagnosis of FG. Originally developed in 2004, it uses a categorical scoring system based on levels of C-reactive protein, white blood cell count, hemoglobin, sodium, creatinine, and glucose. A value of ≥8 indicates high suspicion for FG, 6-7 indicates intermediate suspicion, and <6 indicates low

suspicion (Table 1). This illustrates the importance of obtaining hematological and biochemical laboratory results in a rapid fashion.

The Fournier’s Gangrene Severity Index (FGSI) is a tool created in 1995 used to predict patient mortality. Its underlying premise is based on 9 variables (Table 2). The greater the variation from normal, the greater the mortality in a patient. A patient with a score >9 has an associated 75% chance of mortality, while a score of ≤9 nets a patient a 78% rate of survival.²⁶ Although not a perfect tool, it can aid in predicting survival of patients.

The LRINEC score and FGSI are important tools for diagnosing and predicting mortality in FG; therefore, basic hematological and biochemical labs should be acquired as soon as possible in order to correct electrolyte abnormalities, but also to aid in the diagnosis of FG. These include complete blood count with differential, electrolytes, creatinine, blood urea nitrogen, and blood glucose levels. Arterial blood gas and lactate should be considered as well. It is important to note that the LRINEC score and FGSI do not predict or adjust for immunosuppression.

MEDICAL TREATMENT AND SURGICAL DEBRIDEMENT

Management of FG relies on multimodal therapy, including fluid resuscitation and electrolyte imbalance

Table 1. Laboratory Risk Indicator for Necrotizing Fasciitis

Laboratory values (units)	Score
C-reactive protein, mg/L	
<150	0
≥150	4
White blood cell count, total/mm ³	
<15	0
15-25	1
>25	2
Hemoglobin, g/dL	
>13.5	0
11-13.5	1
<11	2
Sodium, mmol/L	
≥135	0
<135	2
Creatinine, mg/dL	
≤1.60	0
>1.60	2
Glucose, mmol/L	
≤180	0
>180	1

Maximum score is 13; ≥6 raises high suspicion of necrotizing fasciitis, and score ≥8 is strongly predictive of necrotizing fasciitis.

Table 2. Fournier's Gangrene Severity Index

	High abnormal values				Normal	Low abnormal values			
	4	3	2	1	0	1	2	3	4
Temperature, °C	>41	39-40.9		38.5-38.9	36-38.4	34-35.9	32-33.9	30-31.9	<29.9
Heart rate, beats/min	>180	140-179	110-139		70-109		55-69	40-54	<39
Respiratory rate, breaths/min	>50	35-49		25-34	12-24	10-11	6-9		<5
Serum sodium, mmol/L	>180	160-179	155-159	150-154	130-149		120-129	111-119	<110
Serum potassium, mm/L	>7	6-6.9		5.5-5.9	3.5-5.4	3-3.4	2.5-2.9		<2.5
Serum creatinine, mg/dL	>3.5	2-3.4	1.5-1.9		0.6-1.4		<0.6		
Hematocrit, %	>60		50-59.9	46-49.9	30-45.9		20-29.9		<20
White blood cell count, total/ mm ³	>40		20-39.9	15-19.9	3-14.9		1-2.9		<1
Serum bicarbonate, mmol/L	>52	41-51.9		32-40.9	22-31.9		18-21.9	15-17.9	<15

Sum of deviation from normal laboratory values (0-4).

correction, broad-spectrum antibiotic treatment, and rapid and aggressive surgical debridement.

Fluid resuscitation should be initiated immediately as patients often present with sepsis, and electrolyte imbalances and blood glucose levels should be corrected. Bhatnagar et al reported that the majority of FG patients with uncontrolled diabetes presented with diabetic ketoacidosis.⁸ Perry et al found the highest blood glucose in his cohort of patients to be 1,020—a staggering number.²⁵ Managing blood glucose in an acute patient with FG can prove difficult, and it has been suggested that insulin pumps are superior to subcutaneous insulin and should be considered.^{11,27}

A multitude of organisms can cause FG; therefore, empirical broad-spectrum antibiotic therapy should be initiated without delay. This represents a vital cornerstone for the treatment of FG. Triple therapy with penicillins or cephalosporins (gram-positive coverage), aminoglycosides (for gram-negative coverage), and metronidazole or clindamycin (for anaerobes) can be used.²⁸ Monotherapy with broad-spectrum beta-lactams (eg, piperacillin-tazobactam) or carbapenems can also be used.²⁹ Due to the infectious nature of the disease, blood and urine cultures should be collected, including aerobic, anaerobic, and fungal. Once culture results are available, narrowed antibiotic coverage can be considered.

Extensive surgical debridement is considered a cornerstone in the management of FG and preventing its progression to multiorgan failure, sepsis, and death. It has previously been reported that patients forgoing initial debridement face a mortality rate of 100%.²⁶ Zhang et al have reported that debridement should be extensive and should continue until reaching vascularized, bleeding soft tissue, which broad-spectrum antibiotics are able to reach.²⁷ **Debridement should be conducted in a manner that does not involve attempts at salvaging tissue in an attempt to make future reconstruction “easier” or be influenced by the concern for cosmesis, as this can lead to progression of disease due to inadequate resection of nonviable tissue.** Surgeons should have a low threshold for repeat debridement, and wounds should be examined frequently for any acute changes that may suggest progression of infection. On average, patients require between 2 and 4 debridements to achieve infection control.³⁰

TRANSFER OF FG PATIENTS

There has been some debate as to whether FG patients should be transferred from low-volume centers to high-volume centers. Osbun et al compared management of FG patients at high-volume centers to low-volume centers.³¹ They found that low-volume centers had higher rates of orchiectomy when compared to high-volume centers. **More importantly, there was no difference in mortality between these groups; however, delayed transfer from a low-volume to high-volume center was associated with mortality.** This illustrates that a surgeon should not delay early diagnosis, fluid resuscitation, broad-spectrum antibiotics, and surgical debridement in lieu of transferring to a high-volume center. Surgical debridement should be performed as soon as possible in FG patients, as longer time to operating room is associated with higher mortality.

POST-DEBRIDEMENT WOUND MANAGEMENT

Following debridement there are often extensive wounds/defects that need to be managed before reconstruction is initiated. There is no consensus on how to best manage these wounds, and management depends on the depth, size, extent, and contours of the defects. Traditional wound care includes wet-to-dry dressings using different solutions or emollients with a wide variety of active agents such as saline, povidone-iodine, potassium permanganate, Dakin's solution, enzymatic agents for wound cleansing, or polyhexanide.³² Our group typically performed wet-to-dry dressings with one-quarter strength Dakin's solution. These dressings are often easy to change but have the added burden of requiring change at least twice a day.³³

Negative-pressure wound therapy (NPWT) has been used for many decades to remove excess fluid in wounds. Most recently, in 2001 NPWT machines were developed that were covered by Medicare.³⁴ Based on simple principles, NPWT delivers a powerful impact with regard to how wounds are managed and its first reported use in FG patients occurred in 2005. Wound vacuum-assisted closure (VAC) dressings can aid in wound healing via increased perfusion, fibroblast migration, reduction in local edema, and wound edge approximation.³³

A pressure of 50-125 mm Hg with 5 minutes of suction followed by 2 minutes off suction has been recommended for utilizing VAC in FG patients. VAC should only be used after adequate debridement as, by its nature, it can hide progression of disease. Wounds should be evaluated thoroughly during each VAC change, which should occur approximately every 48-72 hours.³³ Downsides to VAC therapy include difficulty maintaining an adequate seal due to the contours of the scrotal and perineal area. Certain novel techniques can assist in obtaining a good seal, though. A recent study reported no leaks with NPWT when a foam dressing was placed in the wound cavity and then primarily suturing wound edges over the foam where possible. This created pockets which kept the foam in place, while also maintaining contact with the wound bed.³⁵ Another concern with VAC use is that the testis can become adhered to the inner thigh, making eventual grafting difficult with unsatisfactory aesthetic results. This can be avoided by applying separate sponges to each individual testis when using a VAC.

Ozturk et al compared VAC (n=5) and conventional wound care (n=5) in FG patients following initial debridement.³⁶ They found both therapies to be equally effective from a wound-healing perspective. Importantly, they also found similar costs associated with both methods. VAC therapy was associated with less pain, lower number of overall dressing changes, and overall better mobility when compared to conventional wound care. Physician time required to perform dressing changes was also reduced in the VAC group. Ultimately, they found that physician and patients had better satisfaction rates with VAC as compared to conventional wound care.³⁶

Bali et al performed a larger study with 33 patients split into a VAC group and a conventional wound care group.³⁷ They noted similar hospital length of stay and need for eventual reconstruction in both groups. They noted a significantly lower re-debridement rate in the VAC group when compared to the conventional wound care group.³⁷

Another important consideration for wound care following debridement is to avoid any fecal contamination. Due to the nature of FG, it may affect areas near the rectum, and stool is a major source of contamination in FG patients. Fecal diversion may be necessary, whether that be a colostomy or ileostomy. However, some argue against the liberal use of colostomy, especially in the elderly, where diversions can become permanent due to anal contracture secondary to disuse. Another concern is patients who have infections tracking up the abdominal wall; diversion can introduce further spread of infection.³⁸ In the same light, FG patients often undergo urinary diversion, as this is often done to reduce the destructive process causing injury to the genitourinary structures as well as reducing contamination. At a minimum, urinary diversion with a suprapubic catheter should be considered in patients with FG to reduce wound contamination.

Hyperbaric oxygen therapy (HBOT) has been used to supplement standard treatment of FG. The principles behind using HBOT in FG rely on its ability to increase oxygen tension in affected tissue, which subsequently is believed to aid in fighting anaerobic bacteria within the tissues affected by FG and improving wound healing.³⁹ A recent meta-analysis identified 10 studies with a total of 657 FG patients, of whom 268 underwent HBOT and 369 did not undergo HBOT. Mortality

rate was significantly lower in those who underwent HBOT, and hospital length of stay was not statistically different between those who underwent HBOT as compared to those who did not. In addition, both groups had a similar number of overall debridements.⁴⁰ The role of HBOT in FG remains controversial. Mindrup et al found that those in their HBOT group actually had a higher mortality than those who did not receive HBOT.⁴¹ On the contrary, Feres et al compared 79 FG patients who underwent HBOT with 118 who underwent standard therapy (wound care, antibiotics, debridement) and found a lower mortality in those treated with HBOT when compared to those who did not receive HBOT.⁴² While HBOT has been explored as an adjunct treatment to surgical debridement for management of FG, its use should be carefully considered, taking each individual patient into account. An absolute contraindication for the use of HBOT is a pneumothorax. Relative contraindications include low seizure threshold, upper respiratory infections, emphysema with CO₂ retention, high fever, and congenital spherocytosis.

Mindrup et al reported a median number of 6 HBOT treatments (range 2 to 26), with an increase in treatments corresponding to more severe illness. Treatment times ranged from 30 to 90 minutes per dive at 2.4 to 3 atmospheres per dive and 1 to 3 dives daily. Cost per treatment on average was \$600 to \$1,300.⁴¹ There is a need for future randomized controlled trials to better characterize the role of HBOT in FG patients.

RECONSTRUCTION

Following early surgical debridement, patients often are left with large genital, perineal, and/or abdominal defects. Debridement defects will eventually need to be managed once the infection has been controlled.

Healing by secondary intention. With smaller genital defects—confined to and involving less than half of the scrotum—healing by secondary intention therapy is possible.⁴³ This is feasible due to the increased elasticity of the scrotum. Healing can be facilitated by loosely approximating skin edges with nonabsorbable suture (eg, nylon), with Akilov et al finding a shorter length of stay with loose wound approximation.⁹

It is important to note, though, that nearly 20% of FG wounds that underwent attempted healing by secondary intention remained open at 6 months.²⁵ While there is unfortunately no correlation between area of defect and time to closure,⁴⁴ those patients healing by secondary intention should be informed that there will be a prolonged time of healing, possible wound contractures, and as a consequence, potentially poor satisfaction with the outcomes.

Given the functional importance of the male genitalia, healing by secondary intention will not be sufficient for many patients and will require some form of reconstructive surgical intervention.

Primary closure with scrotal advancement flaps. **Given the elasticity of the scrotum, larger wounds can often be closed primarily with the use of scrotal advancement flaps.** These flaps offer a good aesthetic result and fulfill the “replace like by like” principle.⁴⁵ Scrotal advancement flaps have been described to repair defects up to 96 cm.^{27,45} Benefits of this reconstruction include durable and good skin quality, elasticity (presence of dartos muscle), and robust blood supply that allows for adequate healing.⁴⁵ It is important that

this closure be tension-free to prevent flap loss and wound edge necrosis.

Skin grafts. In the case of larger defects where primary closure is not feasible, skin grafts can be utilized with adequate results (see Figure). Skin grafts can be used to reconstruct scrotal and penile skin, but do have limitations in the reconstruction of perineal wounds due to poor take secondary to abnormal contours and problematic contractures.⁴⁵ Split-thickness skin grafts are preferred over full-thickness skin grafts to minimize contractures.⁴⁶ Most commonly, split-thickness skin autografts are taken from the lateral thigh, as well as trunk, as these sites are both aesthetically hidden as well as easy to harvest from due to their broad surfaces. Grafts may be meshed to allow for increased scrotal coverage, but when covering the penile shaft, grafts should be left unmeshed in order to optimize cosmesis.⁴⁵ The success of skin grafts is dependent on healthy blood supply of the graft site. It is imperative to have a well-vascularized graft bed to allow for imbibition and inosculation to occur, which, in turn, result in good take of the graft. When skin grafting the testicles, the tunica vaginalis provides a reliable graft bed.⁴⁵ Conversely, when tunica vaginalis is not intact, skin grafting of the testicles will not take. Ultimately, skin grafts are a good reconstructive option with minimal complexity.

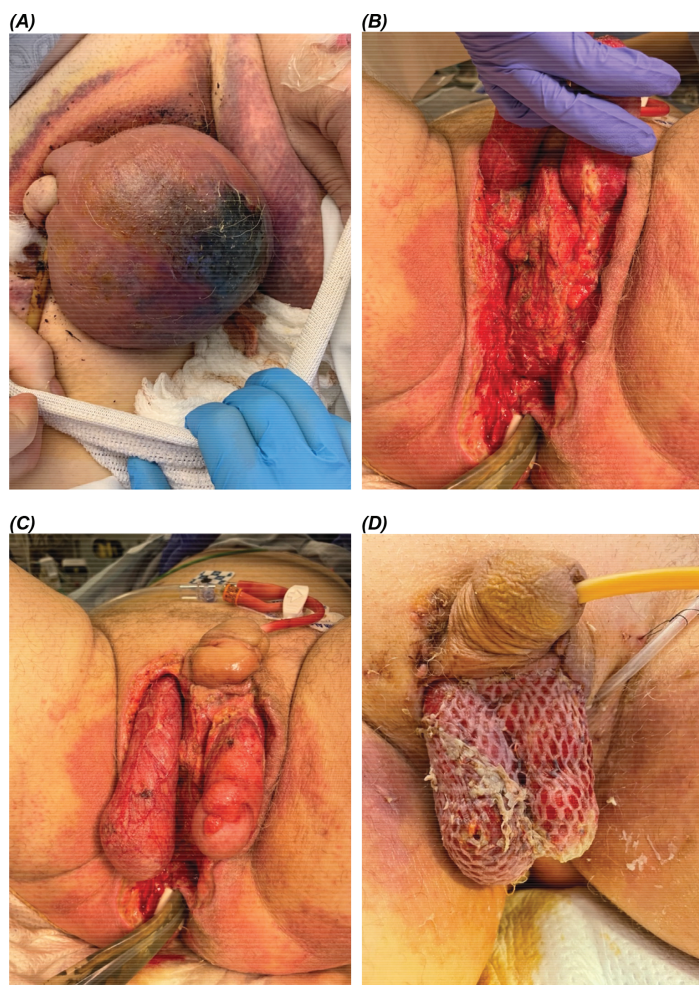


Figure. Fournier's gangrene involving the scrotum and perineum. A, Erythema, edema, and eschar consistent with clinical findings of Fournier's gangrene. B, Debridement extent of perineum. C, Testicles spared during debridement. D, Reconstruction of defect with split-thickness skin graft to recreate scrotum on postoperative day 14.

Flap reconstruction. While much more complex reconstruction techniques, fasciocutaneous and myocutaneous flaps offer superior functionality and cosmesis as well as the ability to fill dead space.⁴³ Examples of such fasciocutaneous flaps include pudendal and medial or lateral thigh flaps.^{12,27} The benefits of these flaps include the presence of a reliable blood supply, preservation of sensation, and minimal donor site morbidity. Meanwhile, gracilis and rectus flaps are examples of myocutaneous flaps that can be used to fill larger/deeper perineal defects.⁴⁷ Another indication for the use of fasciocutaneous flaps is if coverage is needed where the tunica vaginalis is no longer present.⁴⁸

Subcutaneous thigh pouches. Subcutaneous anteromedial thigh pouches have been described as a temporary or permanent solution for placement of the testicles while allowing for further reconstruction of the wound.⁴⁶ Okwudili described 12 cases where testes were placed in thigh pouches to allow primary defect closure later followed by manual massage of the testes back into the scrotal pouch, thus acting as natural tissue expanders.⁴⁹ Despite this, we feel that subcutaneous thigh pouches should be avoided due to poor aesthetics, poor patient satisfaction, chronic testicular pain, and disruption of spermatogenesis caused by elevated testicular temperatures.⁸ Furthermore, with the availability of the VAC, there are exceedingly few instances where thigh pouches are necessary.

Timing of reconstruction. Lastly, the timing of when it is safe to perform reconstruction is unclear, as are the surgical techniques. No studies with high-level evidence have demonstrated the superiority of reconstructive options or approaches. The question of when it is safe to close these FG defects was looked at by Sandberg et al.⁵⁰ Component separation wound closure was performed during initial hospital stay when clinical status was stable, white blood cell count had normalized, and at least 72 hours had passed since initial debridement.⁵⁰ Component separation wound closure is a reconstructive technique that strategically divides subcutaneous and myofascial layers of the abdominal wall in order to achieve tension-free tissue approximation. These authors demonstrated that early component separation wound closure was safe, with postoperative infection complications being similar to those patients who underwent delayed closure or healing by secondary intention.⁵⁰ Furthermore, early wound closure led to a 64% decrease in overall wound convalescence time when controlling for wound size vs healing by secondary intention alone.⁵⁰ Performing early wound closure allowed for less intensive wound care in the hospital and subsequently earlier discharge from the hospital.

CONCLUSION

The treatment for FG consists of immediate surgical debridement and initiation of broad-spectrum antibiotics. Delay in this treatment is associated with increased mortality. Debridement limits are classically influenced not by the concern for cosmesis or function, but by the eventual appearance of vascularized, bleeding soft tissue, which broad-spectrum antibiotics are able to reach. Following stabilization of the patient and resolution of the infection, patients are often left with debridement defects that necessitate reconstruction. VAC-based therapy has improved hospital-based management and reduced the need for subsequent operations. Reconstructive options are

numerous, but choice of reconstructive technique depends on the size of the defect, availability of suitable local tissue, and

depth of resection. Early wound closure may provide shorter convalescence and earlier hospital discharge.

DID YOU KNOW?

- In the acute phase of FG, aggressive fluid resuscitation, broad-spectrum antibiotics, and immediate radical surgical debridement are required. Delay in this treatment is associated with increased mortality.
- Debridement limits should not be influenced by the concern for cosmesis or function, but by the eventual appearance of vascularized, bleeding soft tissue, which broad-spectrum antibiotics are able to reach.
- Secondly, patients are left with debridement defects that will often need definitive reconstruction. Skin grafts and flaps are recommended for reconstruction, depending on the situation.
- Early wound closure may provide shorter convalescence and earlier hospital discharge.

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Study Questions Volume 42 Lesson 19

1. A 56-year-old diabetic man with a progressively decreasing urinary stream presents to the emergency department with a 2-week history of high fevers and chills. Physical examination reveals perineal crepitus, scrotal erythema, and a palpable bladder. The next step is parenteral antibiotics and
 - a. Diverting colostomy
 - b. Hyperbaric oxygen
 - c. Surgical debridement and bladder drainage
 - d. CT-guided percutaneous drainage
2. Progression of a necrotizing soft-tissue infection of the genitalia to the abdominal wall is possible because Colles' fascia is contiguous with
 - a. Camper's fascia
 - b. Fascia lata
 - c. Transversalis fascia
 - d. Scarpa's fascia
3. The most common causative organism(s) of Fournier's gangrene is/are
 - a. Polymicrobial
 - b. *Streptococcus pyogenes*
 - c. *Pseudomonas aeruginosa*
 - d. *Staphylococcus aureus*
4. Worse prognosis in a patient with Fournier's gangrene is associated with
 - a. Higher white blood cell count at time of diagnosis
 - b. Body surface area involvement exceeding 5%
 - c. Urethral involvement
 - d. Presence of crepitus on exam
5. A 65-year-old male undergoes surgical debridement of Fournier's gangrene involving his scrotum. Three days later he is clinically stable and white blood cell count has stabilized. The defect involves less than 50% of the scrotum. The next step is
 - a. Myocutaneous flap
 - b. Hyperbaric oxygen
 - c. Primary closure with scrotal advancement flaps
 - d. Placement of testicles in subcutaneous pouches and delayed closure